

CCP-CNS2026 — Computer Networks

Problem 1: Multi-Site Enterprise Network

Week 2 Status Report

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1. Overview

This week focused on building and verifying the GNS3 network topology for Problem 1. The goal was to get all devices running, configure VLANs, set up inter-VLAN routing using router-on-a-stick, enable DHCP for all three VLANs, and confirm basic connectivity across the network. SSH was also enabled on the router for remote management.

2. Devices Configured

- Cisco 3725 Router (R1) — running IOS 12.4, configured with sub-interfaces for inter-VLAN routing
- GNS3 Ethernet Switch (Switch1) — ports configured with VLAN 10, 20, and 30 access ports and one trunk port
- 3x VPCS (PC1, PC2, PC3) — one in each VLAN, used for testing connectivity

3. What Was Done

VLAN Configuration

Three VLANs were created: VLAN 10 (Sales), VLAN 20 (Engineering), and VLAN 30 (Admin). The switch was configured with one trunk port connecting to the router, and three access ports for each VLAN respectively.

Inter-VLAN Routing (Router-on-a-Stick)

Sub-interfaces were created on FastEthernet0/0 of the router with 802.1Q encapsulation. Each sub-interface was assigned a gateway IP: 192.168.10.1 for Sales, 192.168.20.1 for Engineering, and 192.168.30.1 for Admin.

DHCP Setup

DHCP pools were configured on the router for all three VLANs. All three PCs successfully obtained IP addresses automatically: PC1 got 192.168.10.11, PC2 got 192.168.20.11, and PC3 got 192.168.30.11.

SSH Configuration

SSH version 2 was enabled on the router with an RSA key of 1024 bits. A local admin account was created and the VTY lines were restricted to SSH only.

4. Test Results

- PC1 (Sales) pinged PC2 (Engineering) successfully — inter-VLAN routing confirmed
- PC1 (Sales) pinged PC3 (Admin) successfully

- All three PCs received correct IPs from DHCP with the right gateway
- Router confirmed SSH Enabled - version 2.0 via show ip ssh command

5. Problems Faced

Initially the router image gave an unsupported file extension warning in GNS3. This was resolved by properly adding the Cisco 3725 image through Edit > Preferences > Dynamips > IOS Routers and running the Idle-PC finder to prevent high CPU usage.

The C: drive on the host machine was nearly full (3 GB remaining) so the GNS3 VM and router image were kept on the E: drive to avoid storage issues.

6. Next Steps (Week 3)

- Configure DHCP snooping on the switch to block rogue DHCP servers
- Add ACL to block Sales VLAN from accessing Admin VLAN
- Set up Ubuntu web server in Admin VLAN and Kali Linux attacker in Sales VLAN
- Simulate a DoS attack from Kali and capture traffic using Wireshark
- Add mitigation ACL to block the attacking IP and verify with hit counts